First Terminal Examination - 2082

Class: XI | **Subject: Mathematics (0071)** | **Full Marks: 75**

Time: 3 hrs. | **SET 'A'** | **Pass Marks: 30**

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group 'A'

Rewrite the best alternative of each question in your answer sheet. **[11 × 1 = 11]**

1. The inverse of the statement $\sim p \rightarrow q$ is

   a) $p \rightarrow q$ b) $p \rightarrow \sim q$

   c) $\sim p \rightarrow \sim q$ d) $\sim q \rightarrow p$

2. Which of the following is the statement of De-Morgan's law for logic?

   a) $\sim (p \vee q) \equiv \sim p \wedge \sim q$ b) $\sim (p \Rightarrow q) \equiv p \wedge \sim q$

   c) $\sim (p \wedge q) \equiv \sim p \vee \sim q$ d) Both (a) and (c)

3. If $A \subseteq B$ then $A \cup B =$

   a) B b) A c) $\bar{A}$ d) ϕ

4. For a non-negative real number a, $|x| < a$ implies

   a) $-a \leq x \leq a$ b) $-a < x < a$

   c) $-a > x > a$ d) $-a \geq x \geq a$

5. For $n \in \mathbb{Z}$, the value of $i^{4n+3} =$

   a) 1 b) $-i$ c) -1 d) i

6. If $z = \frac{3+4i}{3-4i}$ then $z\bar{z} =$

   a) 1 b) $\frac{3}{25} + \frac{4}{25}i$

   c) 0 d) $\frac{3}{25} - \frac{4}{25}i$

7. The quadratic equation whose roots are 1 and 2 is

a) $x^2 - 3x - 2 = 0$ b) $x^2 + 3x - 2 = 0$

c) $x^2 - 3x + 2 = 0$ d) $x^2 + 3x + 2 = 0$

8. The quadratic equation $ax^2 + bx - c = 0$ has equal roots if

a) $b^2 < 4ac$ b) $b^2 > ac$

c) $b^2 - 4ac = 0$ d) $b^2 + 4ac = 0$

9. If $\lim_{x \to 3} \frac{x^n - 3^n}{x-3} = 27$ for $n \epsilon z^+$, then $n =$

a) 2 b) 3 c) 4 d) 1

10. If $\lim_{x \to 0} \frac{\sin 5x}{\tan 3x} = \frac{1}{n}$, then $n =$

a) $\frac{3}{5}$ b) $\frac{5}{3}$ c) 0 d) 1

11. $\lim_{x \to 1} \frac{\ln x}{x-1} =$

a) 0 b) -1 c) 1 d) does not exist

Group 'B'

Short Answer Questions. **[8 × 5 = 40]**

12. a) Define tautology. Show that $(p \wedge q) \Rightarrow (p \vee q)$ is a tautology. [1+2]

b) Write the negation of each of the following statements:

i) If $2 \epsilon \mathbb{N}$ then $2 + 1 = 3$. [1]

ii) 7 is even or $5 \times 2 = 10$. [1]

13. a) When two statements become logically equivalent? Show that:

$p \Rightarrow q \equiv \sim p \vee q$ [1+2]

b) If A, B and C are three non-empty subsets of universal set U, prove that: $A \subseteq B \Longrightarrow C - B \subseteq C - A$. [2]

14. a) Define union of two sets in set builder form. If A and B are two non-empty subsets of universal set U then prove that $\overline{A \cup B} = \bar{A} \cap \bar{B}$. [1+2]

b) Find x and y if $5x + 3y + 2ix - 3iy = 4 + 3i$. [2]

15. a) For any two complex numbers z and w, prove that:

$|z + w| \leq |z| + |w|$ [3]

b) Define conjugate of a complex number. Find the conjugate of $\frac{1}{3-2i}$ in the form of $a + ib$. [1+1]

16. a) For what value of 'a' will the equation

$x^2 + (3a - 1)x + 2(a^2 - 1) = 0$ have equal roots? [2]

b) If α and β are the roots of the equation $lx^2 + mx + n = 0; l \neq 0$, then prove that: (i) $\alpha + \beta = -\frac{m}{l}$ and (ii) $\alpha\beta = \frac{n}{l}$ [3]

17. a) If α and β are two roots of the equation $4x^2 + 3x + 7 = 0$ then find $\frac{1}{\alpha} + \frac{1}{\beta}$. [1]

b) Write the condition for existence of limit of a function $f(x)$ at $x = a$. Evaluate: $\lim_{x \to \theta} \frac{x \sin\theta - \theta \sin x}{x - \theta}$ [1+3]

18. a) Evaluate: $\lim_{x \to a} \frac{\sqrt{2x} - \sqrt{3x-a}}{\sqrt{x} - \sqrt{a}}$ [3]

b) If the function $f(x) = \begin{cases} \frac{\sin 3x}{x} & x \neq 0 \\ k & x = 0 \end{cases}$ is continuous at $x = 0$, find the value of k. [2]

19. a) Name any two types of discontinuity of a function. Test the continuity of the function $f(x) = \frac{|x-5|}{x-5}$ at $x = 5$. [1+2]

b) Find the point of discontinuity of $f(x) = \frac{x^2-4x+4}{x-2}$. Also, write the interval for which it is continuous. [1+1]

Group 'C'

Long Answer Questions. **[3 × 8 = 24]**

20. a) Define difference of two non-empty sets in set builder from. For any three non-empty sets A, B and C, prove that:

$A - (B \cap C) = (A - B) \cup (A - C)$ [1+3]

b) If $A = [0, 7]$ and $B = (5, 10)$, find

i) $A \cap B$ ii) $A - B$ [2]

c) For $x, y \epsilon \mathbb{R}$, prove that: $|x - y| \geq |x| - |y|$ [2]

21. a) Write the relation between $z, \bar{z}$ and $|z|$ for any complex number z. If $z^2 = -7 + 24i$, then find the values of z in the form of $a + ib$. [1+3]

b) If a, b, c are rational, then prove that the roots of the equation $(a + c - b)x^2 + 2cx + (b + c - a) = 0$ are rational. [2]

c) If the roots of the quadratic equation $x^2 - 2cx + ab = 0$ are real and unequal, then prove that the roots of

$x^2 - 2(a + b)x + a^2 + b^2 + 2c^2 = 0$ will be imaginary. [2]

22. a) Evaluate: $\lim\limits_{x \to 0} \frac{e^{ax} - e^{bx}}{x}$ [2]

b) Define continuity of a function $f(x)$ at $x = a$. Test the continuity of function

$$f(x) = \begin{cases} \frac{1}{2-x} & for \quad 0 < x < \frac{1}{2} \\ \frac{1}{2} & for \quad x = \frac{1}{2} \\ \frac{1}{3(1-x)} & for \quad \frac{1}{2} < x < 1 \end{cases} \quad \text{at } x = \frac{1}{2}.$$

If it is not continuous, mention the type of discontinuity. If possible, redefine $f(x)$ so as to make it continuous at $x = \frac{1}{2}$. [1+2+1]

c) Evaluate: $\lim\limits_{x \to \infty} (\sqrt{x + a} - \sqrt{x})$ [2]

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